

# Factors Related to University Students' Academic Performance

DSA210 Introduction to Data Science -- Final Report

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Author: Rana Ecem Kılıç

**Abstract.** This project studies factors related to student academic performance and builds machine learning models to predict academic performance levels. The original project dataset is the Higher Education Students Performance Evaluation dataset, which contains 145 observations and education-related features. The initial machine learning version was too limited because it used only one model and mainly reported accuracy. In the final version, the analysis was extended with a separate external UCI Student Performance dataset, multiple machine learning models, cross-validation, classification reports, confusion matrices, and stronger interpretation of the results. The original eight-grade target was also transformed into a three-level target, Low/Medium/High, because the original dataset is too small for reliable eight-class prediction. The best original-dataset model was Gradient Boosting with 0.724 test accuracy. The best external-dataset model with prior grades was Random Forest with 0.885 test accuracy and 0.876 macro F1.

## 1. Introduction and Motivation

Academic performance is one of the most important topics in educational data science because it is connected to student support, course planning, advising, and early warning systems. The motivation of this project is to understand which student-related factors are associated with academic outcomes and to test whether these outcomes can be predicted using data science methods. Instead of only looking at final grades as isolated results, the project analyzes study habits, attendance, previous grade point average, note-taking behavior, and other demographic or educational variables.

The project follows the complete data science pipeline required in DSA210: data collection, data understanding, exploratory data analysis, statistical hypothesis testing, machine learning modeling, model evaluation, interpretation, and reporting. The main research question is: Can student academic performance be explained and predicted using student background and study-related variables?

The first version of the machine learning section was not sufficient because it used only a single Random Forest model and mostly reported only final accuracy. This was not enough for a comparative machine learning analysis. Therefore, the final version was redesigned to include multiple algorithms, stronger metrics, a baseline model, cross-validation, confusion matrices, and a separate external dataset for enrichment and comparison.

## 2. Data Sources and Dataset Diversification

## 2.1 Original dataset

The original dataset used in the project is the Higher Education Students Performance Evaluation dataset. According to the UCI Machine Learning Repository, this dataset was collected from Faculty of Engineering and Faculty of Educational Sciences students in 2019, and its purpose is to predict students' end-of-term performances using machine learning techniques. UCI lists this dataset as a classification dataset with 145 instances and 31 features, and it reports that the dataset has no missing values [1].

In the project repository, the file is stored as `data/student_performance_value.csv`. The repository version contains 145 rows and 32 columns, including the target column `output_grade`. The original output grades are Fail, DD, DC, CC, CB, BB, BA, AA. Important features include weekly study hours, attendance to classes, note-taking behavior, last semester GPA, expected graduation GPA, course ID, scholarship type, and demographic/background variables.

## 2.2 External enrichment dataset

The project guideline states that if a publicly available dataset is used, it should be enriched by another dataset. To satisfy this requirement and to make the machine learning analysis stronger, an external dataset was added as a separate comparative analysis. The external dataset is the UCI Student Performance dataset. UCI describes this dataset as student achievement data from two Portuguese schools, collected using school reports and questionnaires. It provides two course datasets: Mathematics and Portuguese language [2].

The external file added to the repository is `data/external_student_performance_uci.csv`. It contains 1044 course-level observations after combining the Mathematics and Portuguese subsets. The two course subsets were not used to replace the original dataset; they were used as a separate benchmark to test whether similar machine learning methods work on a larger educational performance dataset.

## 2.3 Why the dataset was diversified

The dataset was diversified for three main reasons. First, the original dataset has only 145 observations. This is too small for a stable eight-class grade prediction task. Second, some grade categories have very few examples, which makes it difficult for any classifier to learn minority classes. Third, the project guideline expects public data to be enriched with another data source. The external UCI dataset provides a larger educational dataset with 1044 rows, allowing a stronger comparative machine learning analysis.

Importantly, the original and external datasets were not directly concatenated into one modeling table. Their features are not identical, and they come from different educational contexts. Therefore, the correct method was to analyze them as separate but comparable experiments with the same target idea: academic performance level.

## 3. Project Objective

The objective of the project is to analyze the relationship between student characteristics and academic performance, then build machine learning models that can classify students into broader academic performance groups. The project has two connected goals:

- Explanatory goal: use EDA and hypothesis testing to understand which variables appear related to student grades.

- Predictive goal: use machine learning models to predict academic performance levels and compare different algorithms.

The final version focuses on a three-class target variable instead of exact letter-grade prediction. This makes the problem more realistic for the available data size and still keeps the educational meaning of the target.

## 4. Data Preparation and Target Redefinition

The original target variable had eight separate grade categories. With only 145 rows, predicting eight classes directly produced weak and unstable results. Therefore, the final report uses a broader three-level target:

| Original grade categories | New performance level |
|---------------------------|-----------------------|
| Fail, DD, DC              | Low                   |
| CC, CB                    | Medium                |
| BB, BA, AA                | High                  |

For the external UCI Student Performance dataset, the final grade variable is G3, measured on a 0-20 scale. It was mapped into the same three-class target:

| External dataset G3 score | New performance level |
|---------------------------|-----------------------|
| $G3 < 10$                 | Low                   |
| $10 \leq G3 < 15$         | Medium                |
| $G3 \geq 15$              | High                  |

This transformation is one of the main changes in the final milestone. It was made because broad academic performance groups are more stable, more interpretable, and more suitable for small educational datasets than exact eight-class grade prediction.

## 5. Exploratory Data Analysis

The EDA stage examined dataset shape, column names, data types, missing values, and value counts for important variables. The original dataset did not contain missing values, so the main preprocessing need was categorical handling rather than missing-value repair.

The output grade distribution showed that the most common grade was DD, while some classes such as Fail and CB had much fewer observations. This class imbalance was one reason why exact eight-class prediction was weak. Study-hour analysis showed that many students reported studying less than five hours per week. Attendance was also highly imbalanced because most students reported that they always attended classes.

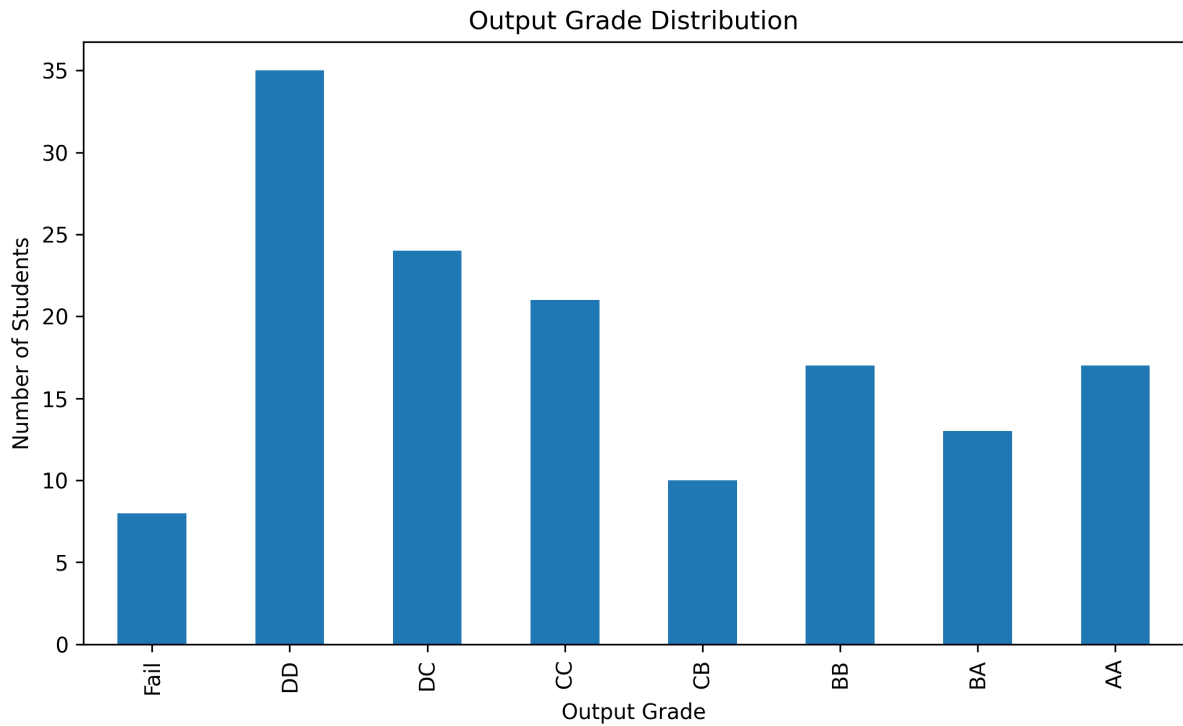


Figure 1. Original output grade distribution.

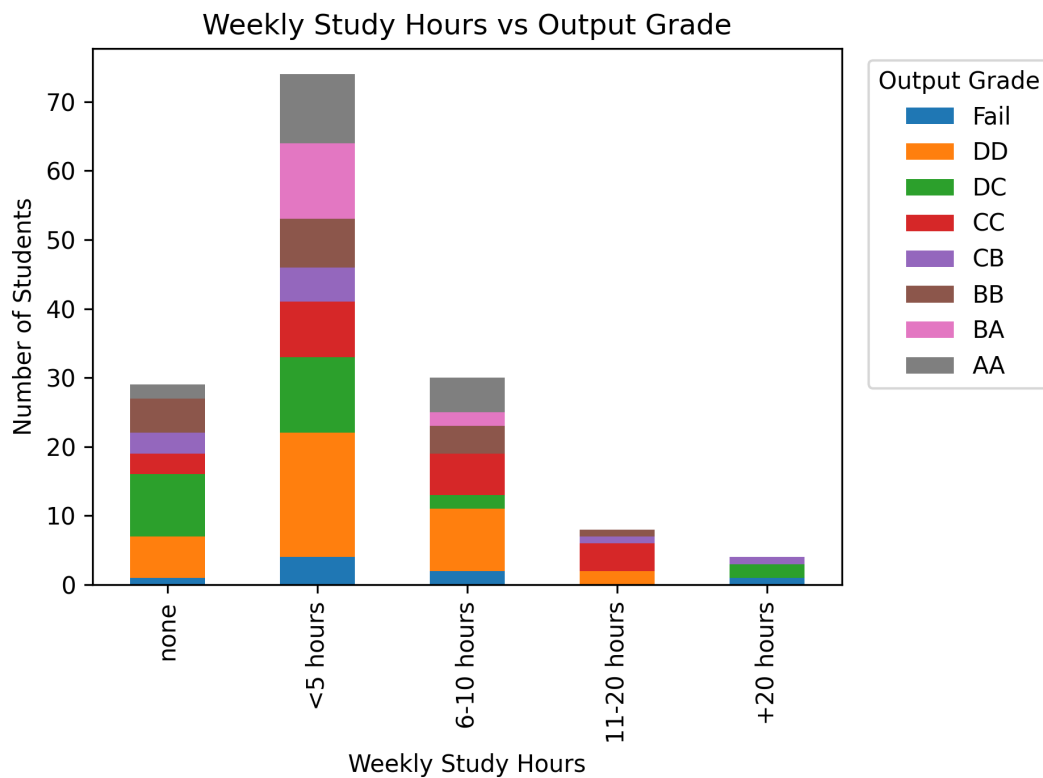


Figure 2. Weekly study hours compared with output grade.

## 6. Hypothesis Testing

The statistical analysis tested whether selected student variables were significantly related to output grade. The tests used chi-square tests of independence for categorical relationships, Spearman correlation for ordered GPA and grade variables, and

Kruskal-Wallis testing for group differences.

| Question                            | Method               | Result                           | Interpretation                     |
|-------------------------------------|----------------------|----------------------------------|------------------------------------|
| Attendance vs. output grade         | Chi-square test      | $p = 0.200408$                   | Not significant at 0.05            |
| Weekly study hours vs. output grade | Chi-square test      | $p = 0.068471$                   | Not significant, but close to 0.05 |
| Last semester GPA vs. output grade  | Spearman correlation | $\rho = 0.3508$ , $p = 0.000015$ | Significant positive relationship  |
| Study-hour groups vs. grade score   | Kruskal-Wallis test  | $p = 0.759686$                   | Not significant at 0.05            |

The clearest statistically significant result was the relationship between last semester GPA and output grade. The Spearman correlation was positive and statistically significant, meaning that students with higher previous GPA tended to have better output grades in this dataset. Attendance and study-hour variables did not reach the 0.05 significance threshold in the current sample, although weekly study hours had a p-value close to 0.05.

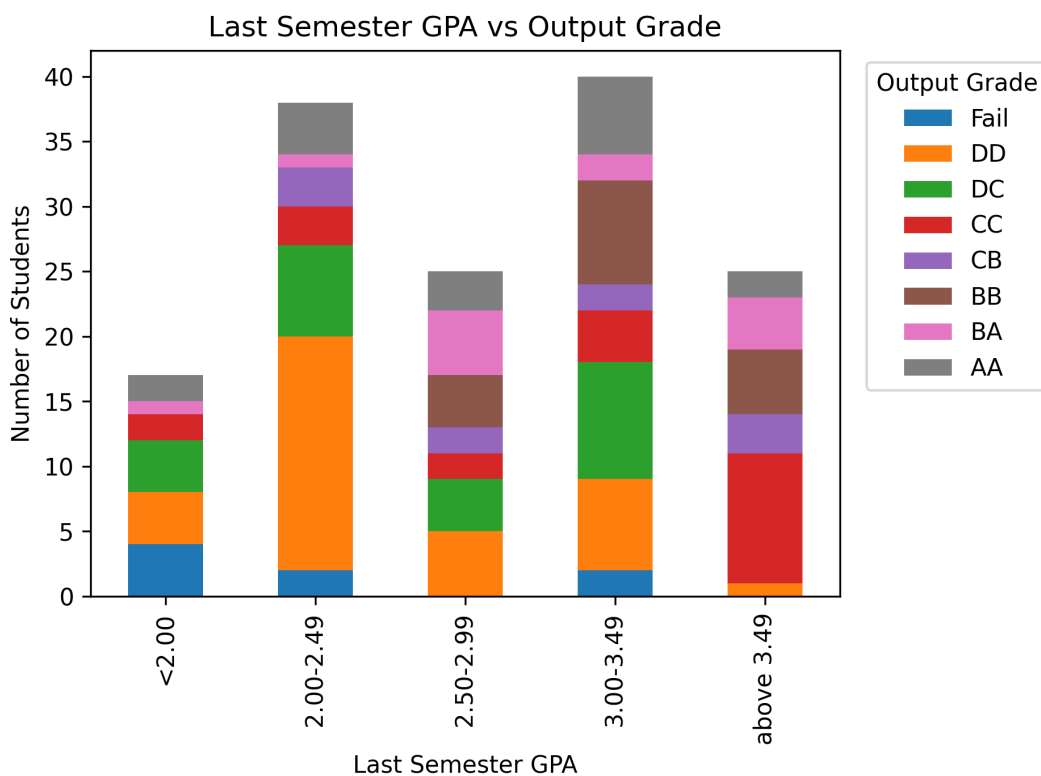


Figure 3. Last semester GPA compared with output grade.

## 7. Machine Learning Methodology

The machine learning stage was rewritten to address the criticism that the earlier version used only one model and did not provide enough evaluation or interpretation. The final ML file is `Milestone2_ML.py`, and a notebook version, `Milestone2_ML.ipynb`, is also included.

The final version evaluates seven model entries: one dummy baseline and six real machine learning classifiers. The evaluated classifiers are Dummy Baseline, Logistic Regression, Linear Support Vector Machine, K-Nearest Neighbors, Decision Tree, Random Forest, and Gradient Boosting.

All experiments used an 80/20 train-test split with stratification, so the Low/Medium/High class proportions were preserved in both train and test sets. The training set was additionally evaluated with three-fold stratified cross-validation. The reported metrics include cross-validation accuracy, cross-validation macro F1, test accuracy, balanced accuracy, macro precision, macro recall, macro F1, weighted F1, classification reports, and confusion matrices.

The preprocessing pipeline handled numeric and categorical variables separately. Numeric columns were imputed with the median and scaled. Categorical columns were imputed with the most frequent category and encoded with one-hot encoding. This is better than using simple label encoding for all categorical variables, because one-hot encoding does not create fake ordinal relationships between categories.

## 8. Machine Learning Experiments

### 8.1 Experiment 1: Original dataset with 3-class target

The original dataset was evaluated using the three-class Low/Medium/High target. The target distribution was 67 Low, 47 High, and 31 Medium observations. The best model was Gradient Boosting, with 0.724 test accuracy and 0.682 macro F1.

| Model               | CV Acc. | CV Macro F1 | Test Acc. | Balanced Acc. | Macro F1 | Weighted F1 |
|---------------------|---------|-------------|-----------|---------------|----------|-------------|
| Gradient Boosting   | 0.707   | 0.629       | 0.724     | 0.682         | 0.682    | 0.729       |
| Decision Tree       | 0.533   | 0.480       | 0.621     | 0.627         | 0.611    | 0.627       |
| Logistic Regression | 0.577   | 0.530       | 0.552     | 0.486         | 0.511    | 0.581       |
| Linear SVM          | 0.585   | 0.514       | 0.517     | 0.453         | 0.475    | 0.542       |
| Random Forest       | 0.569   | 0.504       | 0.586     | 0.474         | 0.465    | 0.556       |
| KNN                 | 0.551   | 0.463       | 0.552     | 0.441         | 0.402    | 0.489       |
| Dummy Baseline      | 0.466   | 0.212       | 0.448     | 0.333         | 0.206    | 0.278       |

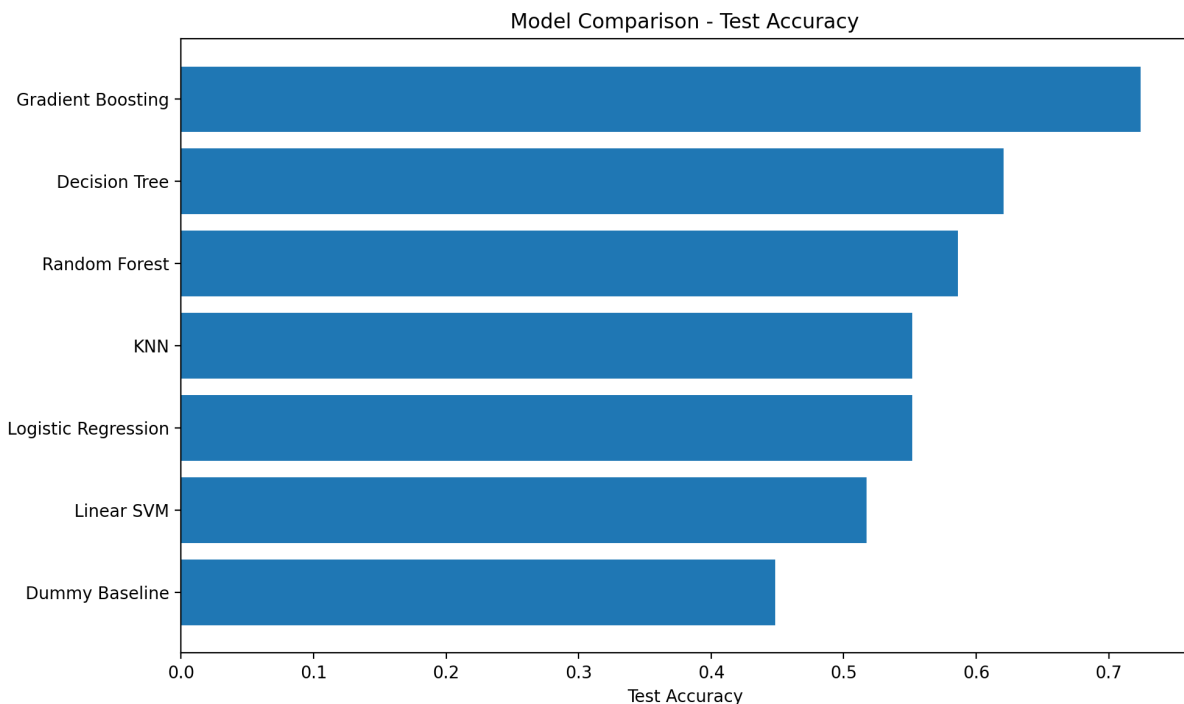


Figure 4. Original dataset model comparison by test accuracy.

## 8.2 Experiment 2: External UCI dataset with prior grades

The external UCI dataset was first evaluated with prior grades G1 and G2 included as input features. These variables represent first-period and second-period grades, so they are naturally strong predictors of the final grade G3. The best model in this setting was Random Forest, with 0.885 test accuracy and 0.876 macro F1.

| Model               | CV Acc. | CV Macro F1 | Test Acc. | Balanced Acc. | Macro F1 | Weighted F1 |
|---------------------|---------|-------------|-----------|---------------|----------|-------------|
| Random Forest       | 0.861   | 0.849       | 0.885     | 0.883         | 0.876    | 0.885       |
| Gradient Boosting   | 0.871   | 0.859       | 0.876     | 0.872         | 0.866    | 0.875       |
| Linear SVM          | 0.831   | 0.820       | 0.861     | 0.905         | 0.860    | 0.862       |
| Decision Tree       | 0.822   | 0.815       | 0.852     | 0.877         | 0.846    | 0.853       |
| Logistic Regression | 0.834   | 0.825       | 0.847     | 0.884         | 0.843    | 0.848       |
| KNN                 | 0.739   | 0.661       | 0.751     | 0.664         | 0.699    | 0.739       |
| Dummy Baseline      | 0.584   | 0.246       | 0.584     | 0.333         | 0.246    | 0.430       |

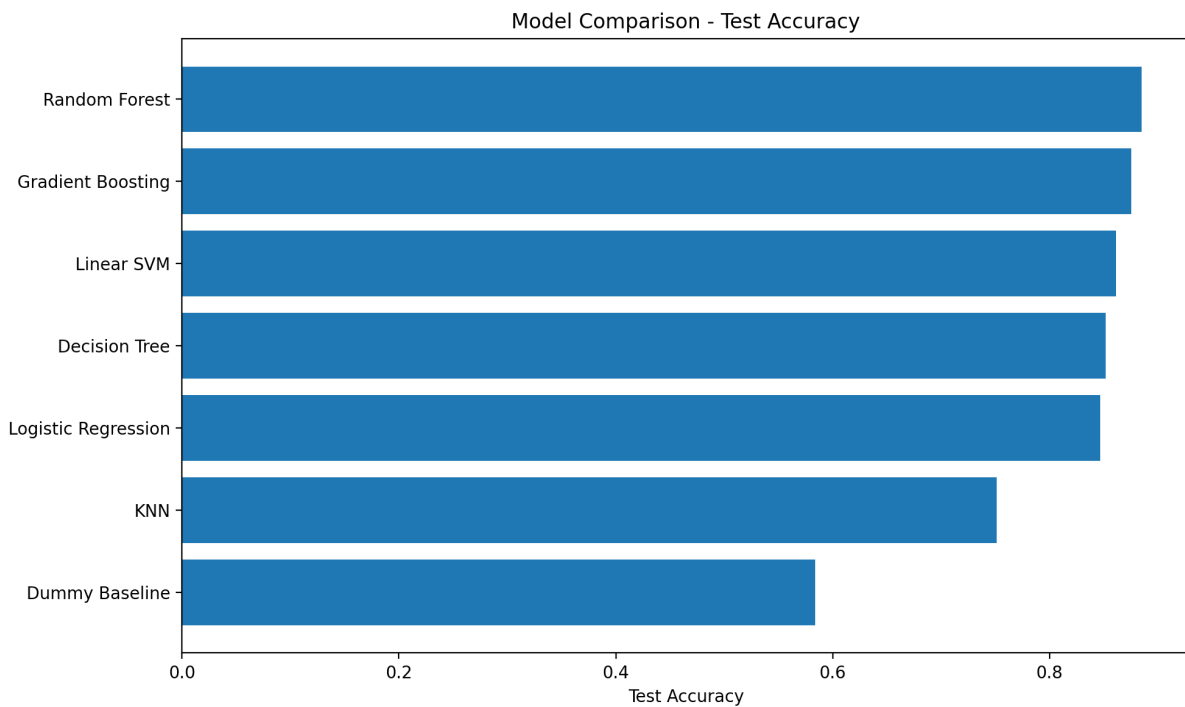


Figure 5. External UCI dataset with prior grades: model comparison by test accuracy.

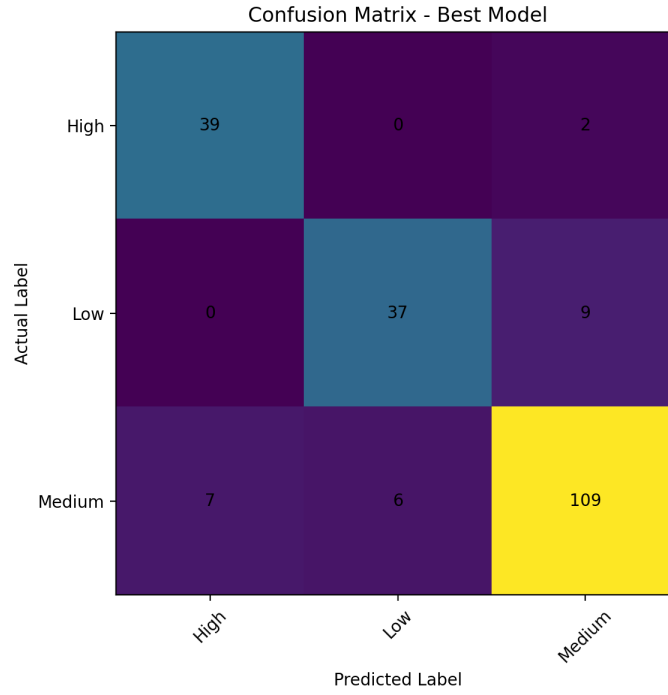


Figure 6. Confusion matrix for the best external model with prior grades.

### 8.3 Experiment 3: External UCI early prediction without G1/G2

To avoid relying only on previous grade information, the external dataset was also tested without G1 and G2. This represents a harder early-prediction scenario, where the model has to predict final performance before earlier course grades are known. In this setting, the best model by macro F1 was Linear SVM, with 0.579 test accuracy and 0.533 macro F1. Although this score is lower, it is more realistic for early intervention because it does not depend on near-final academic signals.

| Model               | CV Acc. | CV Macro F1 | Test Acc. | Balanced Acc. | Macro F1 | Weighted F1 |
|---------------------|---------|-------------|-----------|---------------|----------|-------------|
| Linear SVM          | 0.568   | 0.516       | 0.579     | 0.533         | 0.533    | 0.582       |
| Random Forest       | 0.625   | 0.526       | 0.627     | 0.502         | 0.519    | 0.600       |
| Gradient Boosting   | 0.616   | 0.512       | 0.632     | 0.488         | 0.508    | 0.596       |
| Logistic Regression | 0.513   | 0.502       | 0.483     | 0.563         | 0.487    | 0.480       |
| KNN                 | 0.580   | 0.402       | 0.584     | 0.405         | 0.404    | 0.522       |
| Decision Tree       | 0.419   | 0.421       | 0.364     | 0.473         | 0.371    | 0.338       |
| Dummy Baseline      | 0.584   | 0.246       | 0.584     | 0.333         | 0.246    | 0.430       |



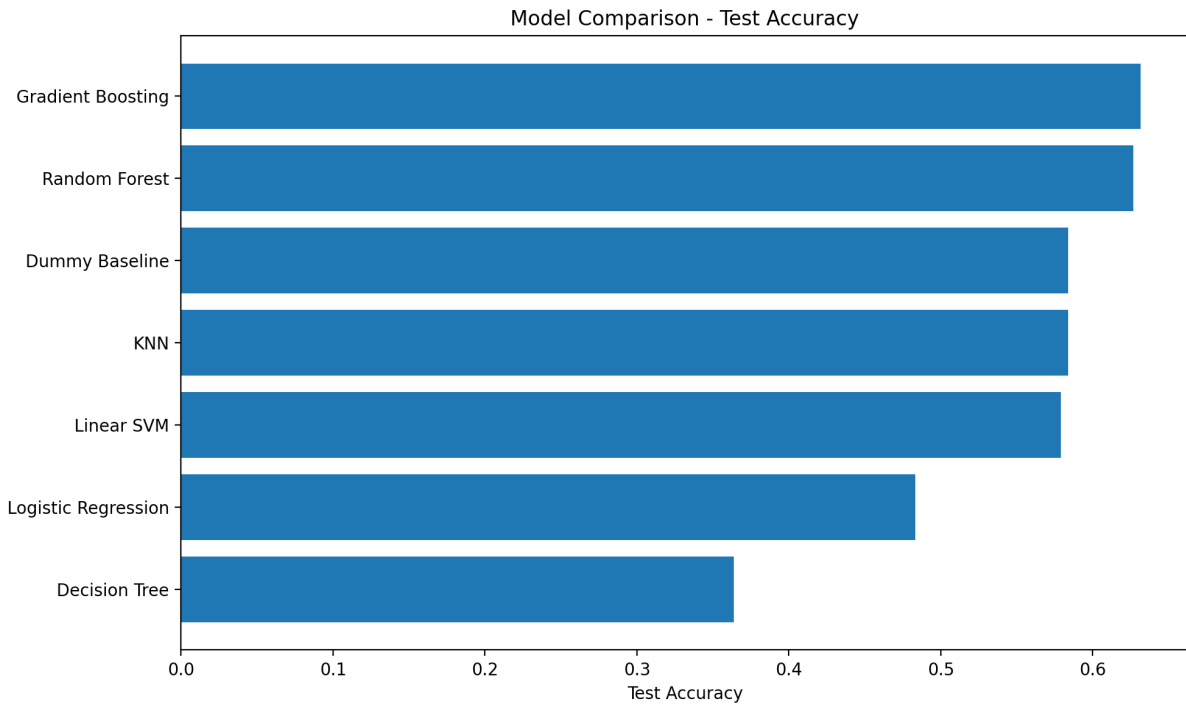


Figure 7. External UCI dataset without prior grades: model comparison by test accuracy.

## 9. Main Results and Interpretation

| Experiment                                           | Best model        | Test accuracy | Macro F1 | Weighted F1 |
|------------------------------------------------------|-------------------|---------------|----------|-------------|
| Original Higher Education dataset: 3-class target    | Gradient Boosting | 0.724         | 0.682    | 0.729       |
| External UCI dataset: with G1/G2 prior grades        | Random Forest     | 0.885         | 0.876    | 0.885       |
| External UCI dataset: early prediction without G1/G2 | Linear SVM        | 0.579         | 0.533    | 0.582       |

The results show that the original dataset can support a reasonable three-class prediction task, but it is not strong enough for exact eight-class grade prediction. The best original-dataset accuracy was 0.724, which is much better than the earlier weak eight-class accuracy. This supports the decision to regroup the target variable.

The strongest overall performance came from the external UCI dataset with prior grades included. Random Forest reached 0.885 test accuracy and 0.876 macro F1. This result is high because previous grades are strong predictors of final grade. This is useful and valid as a comparative ML result, but it should be interpreted carefully because it is not a pure early-warning setting.

The early-prediction external experiment without G1 and G2 had lower performance. This is an important result, not a failure. It shows that predicting academic performance without previous grade information is naturally harder, and that earlier grades contain a large amount of predictive information.

## 10. What Was Changed and Why

- ML section expanded: the previous single-model approach was replaced by a comparative ML analysis with six classifiers and one baseline.

- Evaluation improved: the report now includes accuracy, balanced accuracy, macro F1, weighted F1, classification reports, and confusion matrices instead of only final accuracy.
- Target redefined: the original eight-grade target was grouped into Low/Medium/High because the dataset was too small for stable eight-class classification.
- External dataset added: the UCI Student Performance dataset was added to satisfy the enrichment requirement and to provide a larger comparative benchmark.
- Path handling fixed: the ML script now reads data from the project root using `data/student_performance_value.csv` and `data/external_student_performance_uci.csv`.
- Notebook version added: the ML analysis is available both as a Python script and as a Jupyter notebook.

## 11. Limitations

The main limitation is the small size of the original dataset. With only 145 observations, detailed grade prediction is unstable. A second limitation is class imbalance: some grade categories contain very few samples. A third limitation is that the external UCI dataset comes from a different educational context. It is useful for comparison, but it is not identical to the original higher-education dataset. Therefore, the external dataset was not merged directly with the original dataset.

Another limitation is that high accuracy in the external with-prior-grades setting depends heavily on G1 and G2. These are previous grade variables and are very close to the final grade target. This makes the model strong, but it also means the model is less useful if the goal is very early prediction before any grade information is available.

Finally, the project uses broad performance groups. This improves stability, but it loses some detailed information from the original grade scale.

## 12. Future Work

Future work could improve the project in several ways. The best extension would be collecting more real student observations using the same survey structure as the original dataset. This would allow more reliable exact grade prediction and reduce class imbalance. Another useful extension would be adding feature importance analysis or SHAP explanations to understand which variables drive model predictions. The project could also test resampling methods such as SMOTE, but only on the training set to avoid data leakage.

A stronger future version could build a true early-warning model using only information available before or near the beginning of the semester. This would exclude final or near-final grade variables and focus on attendance, study behavior, background, and early course engagement. Such a model would be more useful for advisors and instructors.

## 13. Ethical Considerations

Student performance prediction must be handled carefully. Educational models should support students, not punish or label them unfairly. Predictions should not be used as final judgments about student ability. They should be interpreted as risk signals that may help instructors provide earlier support. Any real deployment would need privacy protection, responsible data storage, and careful attention to possible bias across student groups.

## 14. Reproducibility

The final repository contains Python scripts, a notebook version, datasets, results, and figures. The main files are:

- Milestone2\_ML.py: final machine learning script
- Milestone2\_ML.ipynb: notebook version of the ML analysis
- data/student\_performance\_value.csv: original project dataset
- data/external\_student\_performance\_uci.csv: external comparative dataset
- results\_ml/: result tables, classification reports, and confusion matrices
- figures\_ml/: generated model comparison and confusion matrix figures
- requirements.txt: Python dependencies

The ML script can be run from the project root with:

```
python Milestone2_ML.py
```

## References

- [1] UCI Machine Learning Repository. Higher Education Students Performance Evaluation. Dataset page states that the dataset was collected from Faculty of Engineering and Faculty of Educational Sciences students in 2019 and contains 145 instances and 31 features. URL: <https://archive.ics.uci.edu/dataset/856/higher%2Beducation%2Bstudents%2Bperformance%2Bevaluation>
- [2] UCI Machine Learning Repository. Student Performance. Dataset page describes student achievement data from two Portuguese schools and provides Mathematics and Portuguese course datasets. URL: <https://archive.ics.uci.edu/dataset/320/student%2Bperformance>